Practice Problems: Probability

Your name here

PSTAT 10

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Instructions

Here are a few problems on probabilty for extra practice. The final exam will be multiple choice(like quizzes) and you will be given a scantron to fill in your answers.

Below are a few problems you need to solve by hand, some by writing the code.

STRATEGY: for getting started with probability problems

Working through probability problems can seem difficult at first. Use this strategy as you think critically about your approach to solve these problems.

- Write down events in words. Let A be the event ....
- Write down the probabilty you want to calculate in math notation using the events you defined. Like: Want to calculate $P(A \cap B)$
- Are these events independent? multiply probabilities or use multiplication rule
- Are these events mutually exclusive? add probabilities or use addition rule
- If the number of ways your event can unfold is long and complicated, look at the complement.

Once you have asked these questions, it'll be a matter of applying the correct formula.

Thinking critically to get answers to these questions is the challenging part of probability. With practice, this will get easier and become second nature!

Exercise 1:

a. Draw 2 cards with replacement. What's the chance of getting two red cards?
b. Draw 2 cards without replacement. What's the chance of getting two red cards?
c. Draw 2 cards without replacement. What's the chance of getting the first red and second not red?

Exercise 2:

The table shows the results of a study in which researchers examined whether athletes participated in a special training program and whether they achieved above-average performance.

	Training Program	No Training	Total
Above Average	28	14	42
Average/Below	32	26	58
Total	60	40	100

Find the probability that...

(a) ... an athlete has above-average performance, given the athlete participated in the training program.

(b) ... an athlete participated in the training program.

(c) ... an athlete has above-average performance, and participated in the training program.

Exercise 3:

A fair six sided die is thrown 4 times. Find the probability that a 5 is obtained each time.

Exercise 4:

You roll a fair die (1-6) twice. Given the sum of the two rolls is equal to 9, what is the probability that both rolls were less than 6?

a. 1/2
b. 1/4
c. 3/8
d. 1/18
e. 5/36

Exercise 5:

What is the expected value of X?

```
##
## x      1.0 5.0 10.0 12.0
## P(X=x) 0.2 0.1  0.3  0.4
```

a. 0.2

b. 1

c. 1.4

d. 28

e. 8.4

Exercise 5:

What scenario can be used to describe a binomial distribution?

a. Sam flips a coin 20 times.

b. Anna records whether today's weather is sunny, rainy, or cloudy for 20 days.

c. Rick shoots basketballs until he makes 20 times.

d. Josh plays video games but his chances of winning the next game increase if he wins the current game.

e. Both A and B

Exercise 7:

Suppose Joe makes it to class on time with probability 0.7. Over a period of 10 classes, what is the probibility he is late to class no more than two times?

a. `dbinom(0,size=10,p=.7)+dbinom(1,size=10,p=.7)+dbinom(2,size=10,p=.7)`

b. `1-pbinom(2,size=10,p-0.7)`

c. `pbinom(8,size=10,p=0.3,lower.tail=FALSE)`

d. `pbinom(2,size=10,p=0.3)`

e. None of the above

Exercise 8:

You are given the following description for the distribution of random variable X: the graph of X is a horizontal straight line from 1 to 10, and 0 otherwise. What is the correct way to calculate the probability that X is between 2.4 and 8.3?

a. `punif(2.4, min=1, max=10)-punif(8.3, min=1, max=10)`

b. `punif(2.4, min=0, max=10)-punif(8.3, min=0, max=10)`

c. `punif(8.3, min=0, max=10)-punif(2.4, min=0, max=10)`

d. `punif(8.3, min=1, max=10)-punif(2.4, min=1, max=10)`

e. `punif(10), min=2.4, max=8.3)-punif(1, min=2.4, max=8.3)`

Exercise 9:

Given the following function, which of the following is true about the output?

```
mean(runif(100, min=1, max=2))
```

a. The output will be around 1.5

b. The output will be around 150

c. The output will be around 100

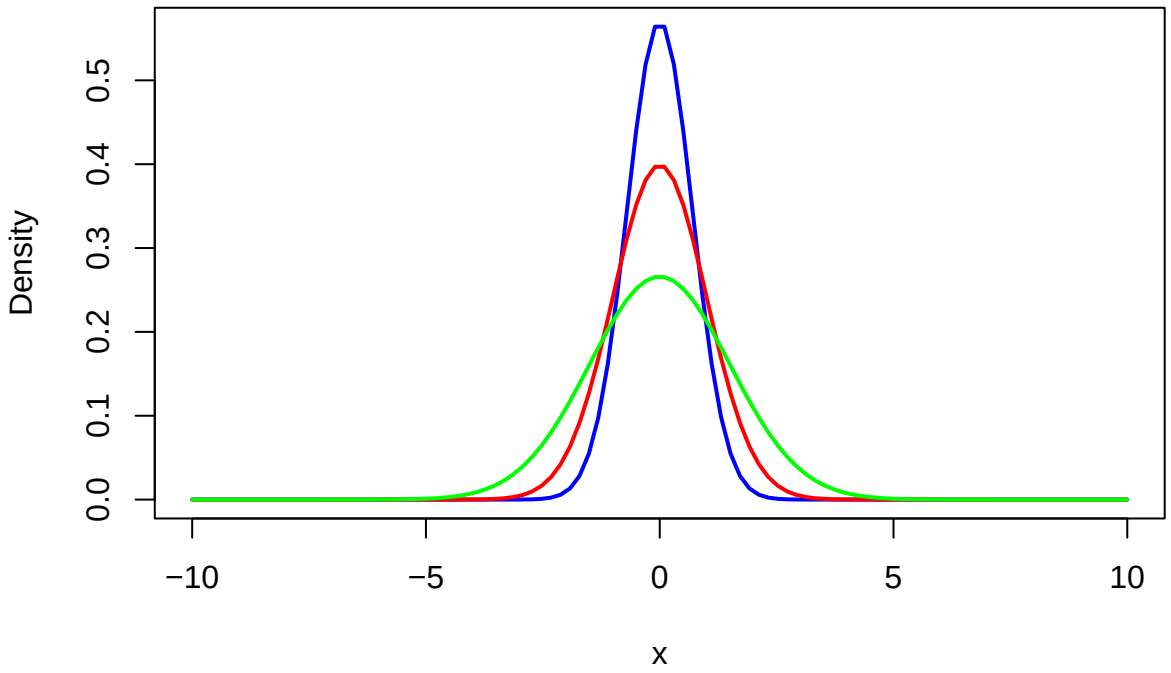
d. The output will be a vector of 100 values ranging from 1 to 2

e. The output will be a singular value ranging anywhere from 1 to 100

Exercise 10:

You are given the following graph with three independent normally distributed random variables. Each of their means are 0, and their respective variances are 0.49, 1, and 2.25 (match each distribution to its correct standard deviation). The green bell curve is selected with probability 1/6, the red bell curve is selected with probability 1/3, and the blue bell curve is selected with probability 1/2.

Suppose you randomly select two bell curves with replacement, then sample a value x from the first bell curve and a value y from the second bell curve. What is the probability on the first pick you select the green bell curve and $x < -1.2$, and on the second pick you select the blue bell curve and $y > 2.1$?



a. `1/6*pnorm(-1.2,0,2.25) * 1/2*pnorm(2.1,0,0.49,lower.tail=FALSE)`

b. `1/2*pnorm(-1.2,0,2.25) * 1/6*pnorm(2.1,0,0.49,lower.tail=FALSE)`

c. `1/6*pnorm(-1.2,0,1.5) * 1/2*pnorm(2.1,0,0.7,lower.tail=FALSE)`

d. `1/2*pnorm(1.2,0,2.25,lower.tail=FALSE) + 1/6*pnorm(2.1,0,0.49,lower.tail=FALSE)`

e. `1/6*pnorm(-1.2,0,1.5) + 1/2*pnorm(2.1,0,0.7,lower.tail=FALSE)`